Sl No	Year	Author and year	Study Location	Study Design	Sample/Population/ context	Method	Analysis type	Phenomena of interest	Model /Classification developed
1	1957	**Fox 1957**	USA	Literature review (Chapter of Book)	Doctors	Narrative	Not applicable	Understanding uncertainty **during training**	Page 208: Uncertainty due to incomplete or imperfect mastry of available knowledge, limitations in current medical knowledge
2	1979	**Light 1979**	USA	Literature review	Medical residents (trainees)	Narrative	Not applicable	Uncertainty and control in **professional training**	Uncertainties Surrounding Instructors, Uncertainties surrounding Knowledge, Uncertaninties of Diagnosis, Uncertaninties of treatment, Uncertaninties of clinical response
3	1983	**Mishel 1983**	USA	Quantitative study	Hospitalised and clinical patients consisting of cardiovascular disease, gastrointestinal conditions, cancer, lupus and symptoms receiving primary treatment	Questionnaire and observation	Cluster analysis (multi- covariate analysis)	Perception of **hospitalised and clinical patients** using Mishel **Uncertainty in illness** scale	Uncertainty clusters for each disease state were identified. Type of uncertainty experienced varied with the type of disease experienced by patients
4	1988	**Mishel 1988**	USA	Literature review	Studies from nursing and related fields are used for discussion	Narrative	Not applicable	Taxonomy of **uncertainty in illness** (people perception and how they respond)	Model of perceived uncertainty in illness is developed. myriad sources of uncertainty in life are re¯ected in the statement: `The experience of uncertainty is pervasive in human existence. Perception of being unable to assign probabilities for outcomes.'
5	1989	**Smithson 1989**	USA	Literature review (Chapter of Book)	General	Narrative	Not applicable	Ignorance and Uncertainty	"available information is limited to or characterized by probability (the likelihood of a future event), ambiguity (a multiplicity of possible states for a single concept or event), or vagueness (a multiplicity of possible values on a continuum".
6	1991	**Beresford 1991**	Canada	Qualitative study	Clinicians (Physicians) in a variety of healthcare settings	Interviews	Descriptive	Uncertainty shaping **medical decissions (allocation of medical resources)**	Technical , Personal, Conceptual uncertainties were identified. (From the representative comments we could categorise further uncertainties)
7	1995	**Sheer 1995**	USA	Quantitative study	Cardiac patients at the waiting room of cardiology clinic	Survey (pre-visit and post visit questionnaires	Pearsons correlation and Wilcoxan tests	Uncertainties in **physician-patient interactions**	Pre-visit uncertainty, Illness uncertainties, relational uncertainties, Medical setting uncertainty ; Post-visit uncertainty
8	1998	**Babrow 1998**	Mexico	Literature review	Uncertainty experienced by physicians and by extention patients	Narrative	Not applicable	Systematic analysis of many meanings of uncertainty in **health and illness experiences**	5 principal forms – Complexity, Qualities of information, probability, structure of information, lay epistemology
9	2002	**Penrod 2002**	USA	Literature review	Nursing	Review (with systematic search)	Refiend concept analysis	to analyse the conceptual maturity of uncertainty with the scientific literature on uncertainty and to refine the concept	uncertainty related to prognosis, course of treatment, choice of provider and family relationships.
10	2004	**Begun 2004**	USA	Literature review	Organisational environement	Narrative	Not applicable	Uncertainty in health care environment.	A framework for analysing uncertainty in health care organisational environments is given. Complexity and dynamism are reproted as the main components of environmental uncertainty
11	2008	**Farnan 2008**	USA	Qualitative study	Internal medicine Residents	Interviews of residents reporting critical incidents	Thematic analysis	Resident uncertainty in critical incident clinical decision making and patient care	Technical uncertainty (procedural skills, knowledge of indications); Conceptual uncertainty (care transitions, diagnostic decision making and management conflict) and Personal uncertainty (goals of care). (From the representative comments we could categorise further uncertainties)
12	2008	**Kasper 2008**	Germany	Qualitative study	Cancer patients	In- depth interviews	Grounded theory	Taxonomy of uncertainty experienced by cancer patients	8 categories were identified. (1) disease-related issues (prognosis/diagnosis, treatment), (2) risk communication issues (deciphering information, role in the medical dyad, physician's trustability) and (3) aspects of coping with life considering the disease (mastering requirements, social integration, causal attribution).
13	2011	**Brown 2011**	UK	Qualitative study	Data was collected from three different documents- 1. FAD documents which provide a rather disembodied and bureaucratically 'sanitised' account of the decision-making process. 2. the 'Report of Threshold Workshop' (NICE 2009) enables deeper insights into how the threshold is applied within NICE appraisal committees 3. the transcript of a relatively candid interview given by Sir Ian Kennedy (2009), regarding his research into the way NICE deals with innovative health technologies.	Content analysis of a selection of NICE documents	Narrative	A framework is developed which posits the various forms of uncertainty, and responses to these, as potential conduits of regulatory bias—in need of further research.	Epistemic ,Procedural , Interpersonal were identified. (From the representative comments we could categorise further uncertainties)
14	2011	**Han 2011**	USA	Literature reivew	Health care	Narrative review	Synthesizing diverse theoretical and empirical literature from the fields of communication, decision science, engineering, health services research, and psychology, and developing a new integrative conceptual taxonomy of uncertainty	Taxonomy of uncertainty in health care	Uncertainty classified based on sources, issues and loci
15	2011	**Djulbergovic 2011**	USA	Literature review (Chapter of Book)	Uncertainty in medicine, both in clinical practice and in clinical research.	Narrative	Narrative	Authors perception on uncertainty in clincal research and clinical practice	Epistemic , subjective, linguistic uncertainty and Probability
16	2012	**Maffei 2012**	USA	Qualitative study	Male subjects (patients) from philadelphia community	Interviews	Content analysis	Uncertanity in Shared Decision-making in Medically Uncertain Situations	Past experiences, Individual personality, Relationships with patient and physician
17	2012	**Soliman 2021**	USA	Case study	Expert opinion on uncertainties in non-compaction cardiomyopathy	Case study based	Narrative	Diagnostic uncertainties in non compaction cardiomyopathy	Genetic uncertainities, Dignostic uncertainites
18	2013	**Cristancho 2013**	Cannada	Qualitative study	Cases drawn from seven staff surgeons during 26 surgu=ical cases	Observations and interviews	Template and inductive analysis	Uncertanity in clinical setting (Surgicla cases)	Incomplete information, Patient related charecteristics (Abnormal anatomy), External factors- Equipment issues , Staffing, time pressure

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19	2013	**Greenhalgh 2013**	UK	Case study	Differnet patient cases - having a breast lump, itchy rash, painful hip, wax in the ears	Case studies	Narrative	Taxonomy of uncertainties in clinical methods	Probability (Bayesian), Open-endedness of story ; Environment; Network of people, Ethical. (From the representative comments we could categorise further uncertainties)
20	2014	**Kerr 2014**	USA	Qualitative study	Parents of children with complex chronic illnesses	Open ended questionnaire and observation	Constant comparison analysis	Parental uncertainties surrounding an 'Orphan' Illness (not belonging to one medical speciality)	Normalisation (probability), Inadequate information communicated to patient, Multiple conflicting opinions, Who is giving the information; Proxy uncertainty (Taking decisions on behalf of someone)
21	2015	**Hamui-Sutton 2015**	Mexico	Mixed- Method (only qulaitative data used for the review)	Medical residents	Combination of interview and questionnaire (Only qualitative component used to develop typology of uncertainties)	Content validation of the critical incident reprots	Uncertainty in ctitical incidents in medical residents	Conceptual- Inability to apply abstract knowledge in concrete situations, Technical: Lack of theoretical information, Ignorance for guiding action, Physician inability to communicate; Systemic – due to lack of technological, technical, material and human resources; Ethical Values and socio cultural codes of society/institution / person
22	2015	**Wray 2015**	USA	Literature review	Uncertainty physicians face during diagnosis, treatment planning and in prognostic decision making	Information from fields of sociology, anthropology, psychology and authors experience	Not applicable	Medical Uncertainty in diagnosis, prognosis, and treatment	Informational uncertainty , Intrinsivc uncertainty
23	2016	**Danczak 2016**	USA	Qualitative study	GP associates in training	Focus group discussion	Thematic analysis	GP associates in training and their experiences of uncertainty	Analysing, networking, negotiating, team working (From the representative comments we could categorise further uncertainties)
24	2017	**Etkind 2017**	UK	Qualitative study	Interviews with 30 patients with heart failure, COPD , renal disease , cancer and liver disease	Secondary analysis of qualitative interviews from six studies including patients with heart failure, chronic obstructive pulmonary	Thematic analysis	understand patient experiences of uncertainty in advanced illness and develop a typology of patients' responses	Patient information needs and preferences Complexity; Illness course, Seriousness; Patient engagement, Temporal focus
25	2017	**Han 2017**	USA	Qualitative study	Leaders in clinical genomics field	Empirically informed by a synthesis of insights from open-ended qualitative interviews undertaken with leaders in the clinical genomics field.	Thematic analysis	Uncertainties in clinical genome sequencing	Source , isue loci similar to original to Hans 2011 taxonomy but with additional domains is given
26	2017	**Lau 2017**	Denmark	Case study	Patients with high cholesterol taking statins	Case studies and ethnographic interviews	Narrative	Uncertainty in patients taking statins	Epistemological,Relational,Aleatoric (From the representative comments we could categorise further uncertainties)
27	2017	**Ueki 2017**	Japan	Qualitative study	Maternal Uncertainty about Infants' Hospitalization	unstructured interviews	Content analysis	Maternal Uncertainty about Infants' Hospitalization for Acute Childhood Illness	five categories: ambiguity about the severity of the illness, unpredictability of the course of the illness, discrepancy of the judgement with healthcare professionals, lack of information about the causes of the illness, and ambiguous appropriateness of management.
28	2018	**Kang 2018**	USA	Case study	When radiologists encounter incidental findings	Case study based	To support decision making for the management of incidental findings, major sources of uncertainty in management pathways can be mapped and analyzed using mathematical models.	Uncertainty in managing incidental findings	Sources: Scenario uncertainty, Statistical uncertainty; Loci:Health care system , Provider, Patient ; and several issues like diagnostic possibilites , test tolerance etc
29	2020	**Hong 2020**	Singapore	Literature review	Communicating uncertainties to patients	Scoping Review	Thematic analysis	Uncertainties in cancer related genertic risk communication	Understanding information, future risks managing information; Impact on family and life; Utility (value) of information Patients feeling before test
30	2020	**Kelly 2020**	UK	Case study	Emergency crisis (Pandemic)	Analysis of Zika virus case study	Narrative	Uncertainty in medical emergencies	Global uncertainty, Public health, clinical (Taking decisions on behalf of someone)
31	2020	**Lee 2020**	New Zealand	Literature review	Medical students	Scoping Review	Content and inductive analyses	Uncertainty in medical education	Lack of knowledge, Nature of knowledge, Application of knowledge; Systems uncertainty; Relationships
32	2020	**Pomare 2020**	Australia	Literature review	Health care professionals	Scoping review	Framework synthesis	Uncertainty in complex healthcare settings	Hans classification is modified by adding ethical and system added and integrated model presented
33	2021	**Brondani 2021**	Canada	Qualitative study	Frontline oral health care workers, including dentists, dental hygienists, certified dental assistants, and administrative staff	Remote semi-structured interviews	Thematic analysis	Uncertainties, as experienced by members of the oral health care workforce in British Columbia	Uncertainties surrounding COVID-19, uncertainties surrounding patient care, uncertainties toward personal lives and infectiousness